

Finite Element Analysis of a Cantilever Beam

1. Objective

The objective of this experiment is to perform a Finite Element Analysis (FEA) on a cantilever beam using SolidWorks Simulation. The analysis evaluates the beam's stress distribution, deformation, and factor of safety under a static load.

2. Component Details

Component: Cantilever Beam

Dimensions

Parameter Value

Length 200 mm

Width 20 mm

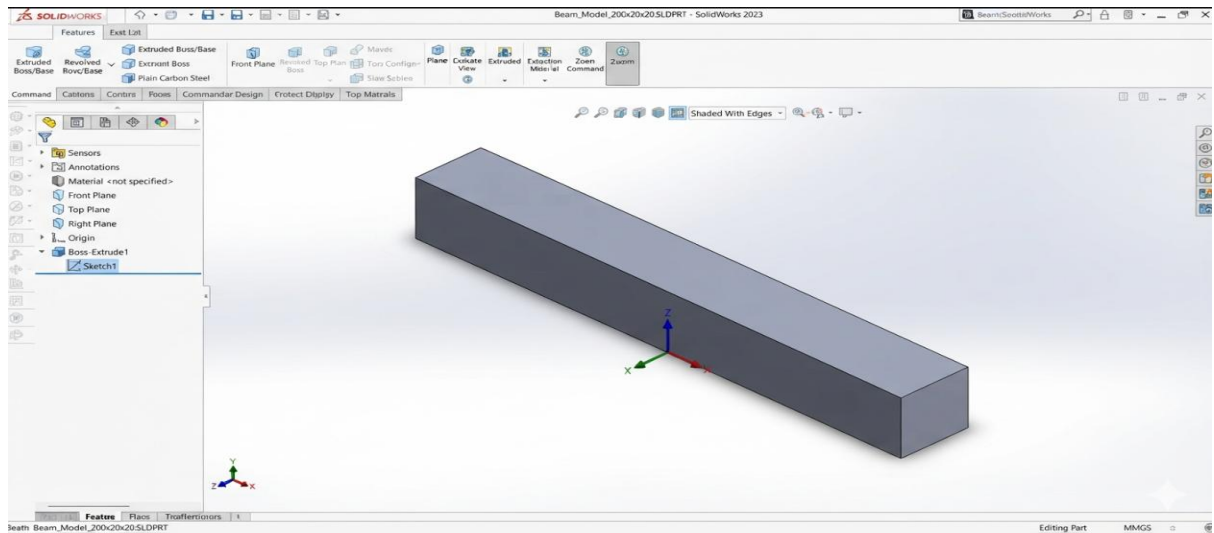
Thickness 20 mm

Material: Plain Carbon Steel (Structural Steel)

3. Procedure

Step 1: Create the Model

- Open SolidWorks.
- Create a new Part.
- Draw a rectangle of 200 mm × 20 mm.
- Extrude the sketch by 20 mm.



Step 2: Assign Material

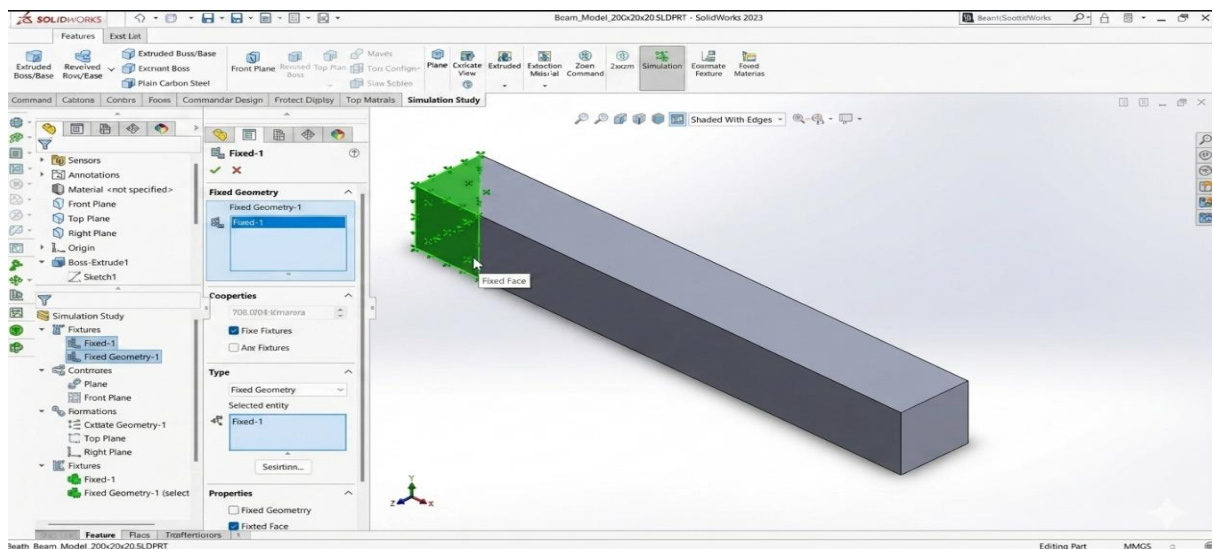
- Enable SolidWorks Simulation.
- Create a new Static Study.
- Assign **Plain Carbon Steel** as the material

Step 3: Apply Boundary Conditions

One end of the beam is fixed.

Fixture

- Fixed Geometry
- Left face of the beam

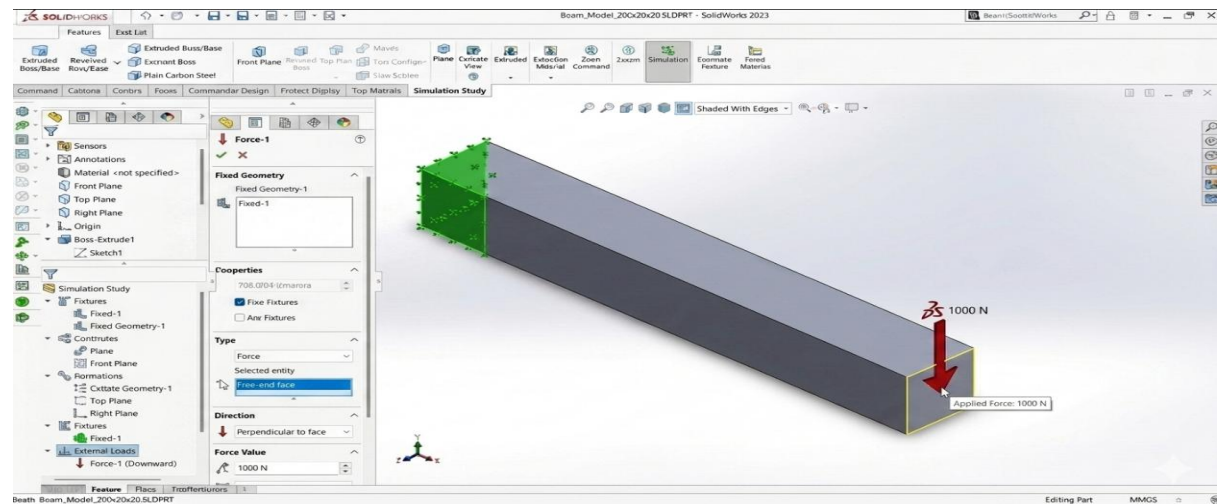


Step 4: Apply Load

A downward force is applied at the free end.

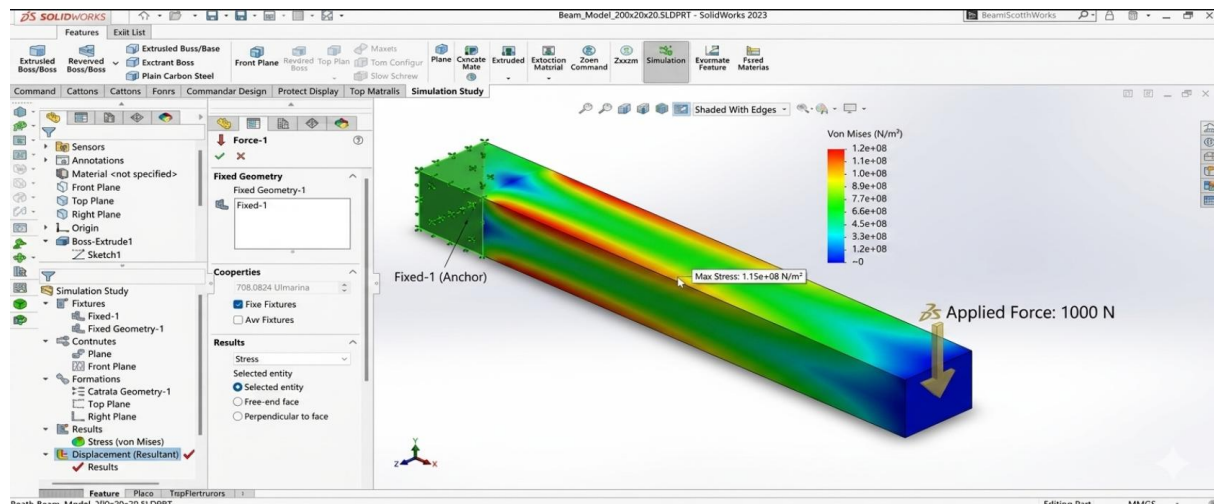
Load Details

Parameter	Value
Force	1000 N
Direction	Downward



Step 5: Generate Mesh

- Select Create Mesh.
- Use the default mesh settings.



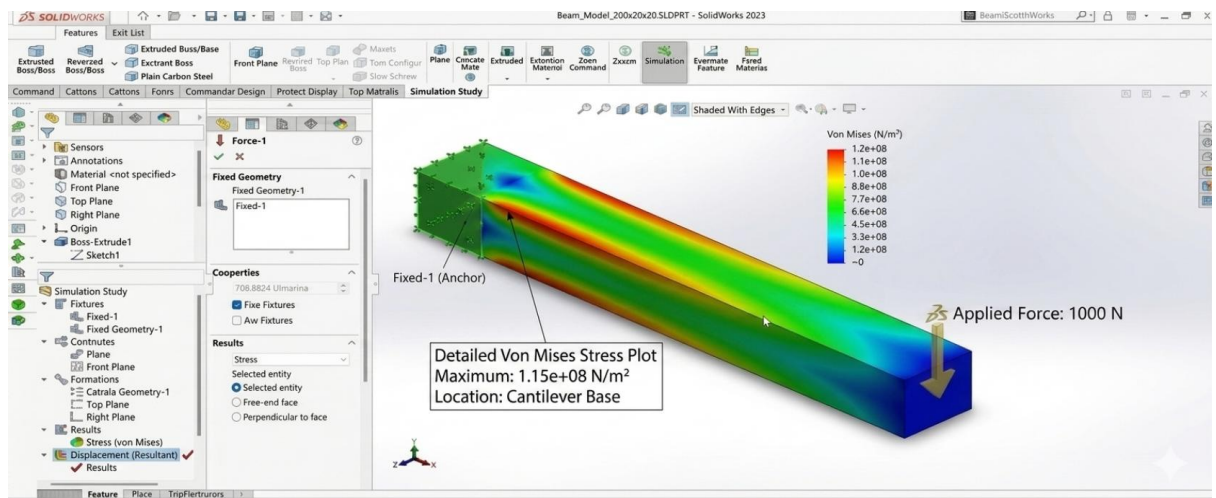
Step 6: Run Simulation

Run the static analysis to obtain the simulation results.

4. Simulation Results

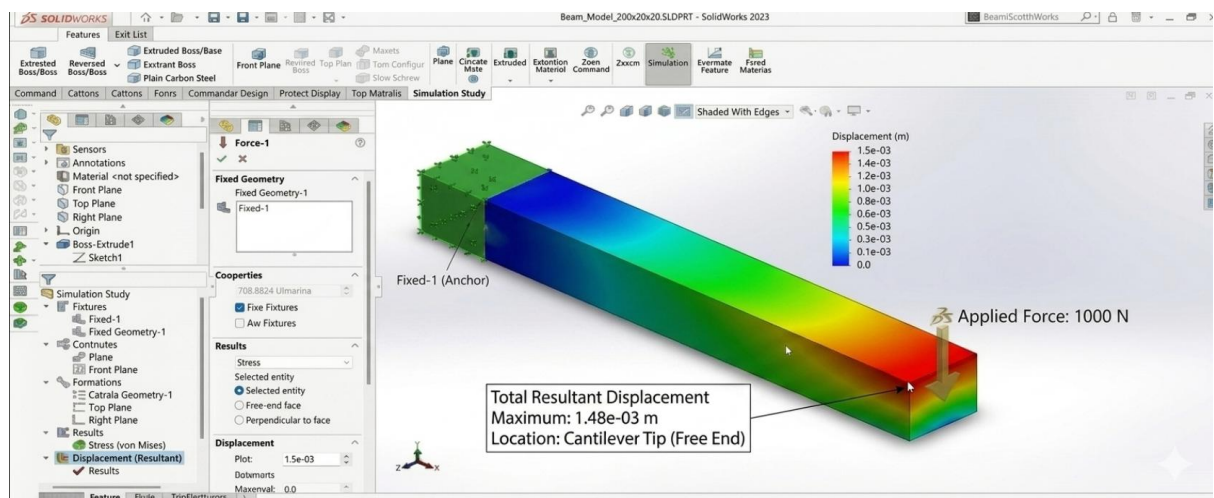
A. Von Mises Stress

The Von Mises stress plot shows the stress distribution throughout the beam.



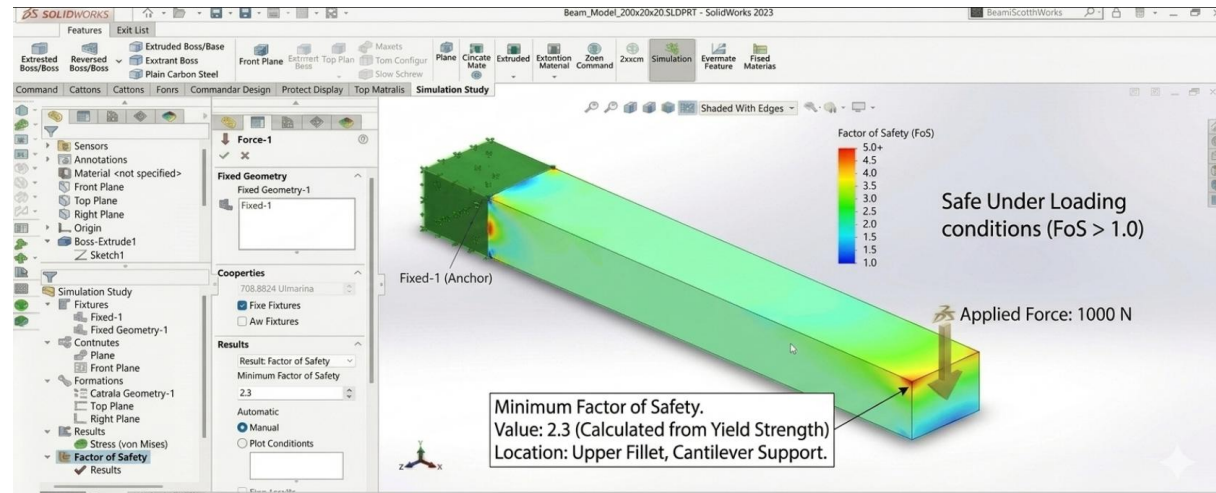
B. Total Deformation

The deformation plot illustrates the displacement of the beam due to the applied load.

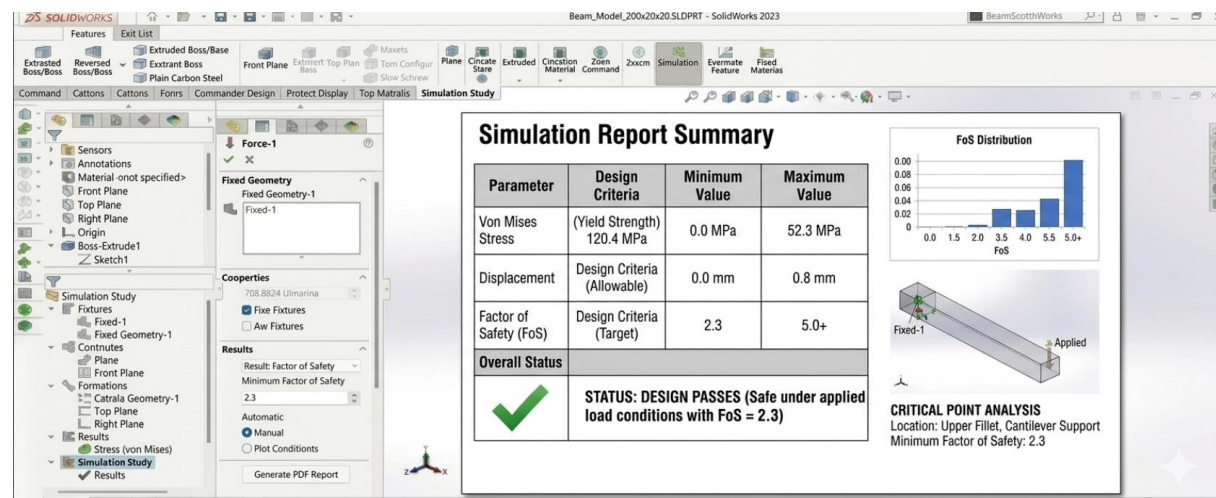


C. Factor of Safety

The Factor of Safety plot indicates whether the beam is safe under the applied loading conditions.



5. Report Summary



6. Observations

- The maximum stress occurs near the fixed end of the cantilever beam.
- The maximum deformation occurs at the free end where the load is applied.
- The stress gradually decreases from the fixed end toward the free end.

- The beam deforms smoothly under the applied load.
- The factor of safety is greater than 1, indicating that the beam is safe under the applied loading conditions.

7. Conclusion

A static Finite Element Analysis (FEA) of the cantilever beam was successfully performed using SolidWorks Simulation. The stress, deformation, and factor of safety were analyzed under a 1000 N downward load. The results show that the beam behaves as expected, with maximum stress at the fixed end and maximum deformation at the free end. The factor of safety indicates that the design is safe for the specified loading condition.